Stage C: pipeline validation (code-level checks)

FINAL_V1 - Parkinson's disease detection from speech

Protocol FINAL_V1 Status: **COMPLETED** Generated: 2026-10-20T09:41:57Z

What was validated

- 25 automatic checks on synthetic features and the audited manifests; result: ALL PASSED.
- Instrumented run proved that every fit uses only rows of its own training partition, that tuning never touches outer-test rows or speakers, and that feature selection sees training speakers only.
- A null control (random features, true labels) gave chance-level pooled AUROC and a positive control gave near-perfect AUROC, so the nested loop neither invents nor loses signal.
- A deliberately corrupted prediction file (test speaker also in training) failed three integrity gates and was quarantined, not repaired.
- Resume logic reuses completed runs. All 78 frozen hyperparameter configurations fit and return probabilities in [0,1].

Separation of smoke tests

- Smoke outputs live in tests/tmp_smoke and are labelled SMOKE_TESTS_NOT_EXPERIMENT_RESULTS; they are excluded from every metric and table of the study.

Table 1: Check results

Test	Description	Result
T01	label encoding: PD folder -¿ 1, elderly folder -¿ 0	PASS
T01b	positive class is PD (label 1) and counts 25/22 speakers	PASS
T02	outer folds speaker-disjoint and cover all 47 speakers	PASS
T02b	every outer test fold contains both classes	PASS
T02c	every recording sits in its speaker's fold	PASS
T02d	inner folds speaker-disjoint, inside the training partition, both classes in each inner validat	PASS
T02e	split is reproducible (same seed -¿ same assignment)	PASS
T03	sample_ids unique, no duplicated source audio, no speaker in two groups	PASS
T04	acoustic matrix aligned by sample_id, 40 columns, no missing	PASS
T04b	acoustic file sample_id order equals manifest	PASS
T05	metrics toy example (TP2 FN1 TN1 FP1)	PASS
T05b	average precision toy	PASS
T05c	speaker score is the mean of recording probabilities	PASS
T06	changing one test row leaves the other test predictions unchanged (no test statistics in scalin	PASS
T07	outer-test rows appear only as the prediction target of the single final fit	PASS
T07b	every tuning fit trains and validates on outer-training rows only; inner train/validation disjo	PASS
T07c	feature selection never sees an outer-test speaker (4 selections: 3 inner + 1 final)	PASS
T07d	the selector for the final model used the whole outer-training partition	PASS
T08	all 78 frozen configurations fit and return finite probabilities in [0,1]	PASS
T09	null control (random features, true labels): pooled speaker AUROC near chance	PASS
T09b	positive control (label-informative feature): pooled AUROC high	PASS
T10	checkpoint/resume: completed run is reused, not recomputed	PASS
T11	corrupted prediction file fails the gates and is quarantined (not repaired)	PASS
T12	hashing is stable; registry has 280 fold runs, 56 condition rows and the blocked FWL row	PASS
T12b	frozen hashes equal the files on disk	PASS